

CS360 Introduction to Database - Midterm Problem Sheet

- **Date & Time:** April 16 (Wednesday), 13:00 ~ 15:00
 - **Separate answer sheet:** This is a question sheet, and all grading will be based on the answers on the separate answer sheet only. Write clear answers on the answer sheet please.
 - **Name on answer sheet:** You must write your name on the answer sheet. (1 point is given)
 - **No Electronic Devices:** The use of any electronic devices is strictly prohibited.
 - **Questions During the Exam:** If you have any questions, please raise your hand.
 - **Deduction Policy:** The score for each Problem is non-negative. (i.e., $\max(\text{sum}(\text{score}), 0)$)
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Problem 1 (6 points)

Determine whether the following statements are True (T) or False (F).

(1 point for each correct, -1 point for each incorrect)

1. Every relation in 4NF is also in BCNF.
2. R is in BCNF if no nonprime attribute of R is transitively dependent on the primary key of R .
3. In database design, a multiway relationship is generally recommended.
4. In set-based relational algebra, say $|R|$ is the cardinality of R , then $|R - S| = |R| - |S|$.
5. Most commercial DBMS's implement relations that are bags.
6. Safe single datalog rules can be expressed as relational algebraic operations such as join, selection, and projection.

Problem 2 (8 points)

Determine whether the following statements are True (T) or False (F).

(2 points for each correct, -2 points for each incorrect)

1. An arbitrary 3-column relation can have at most three different candidate keys.
2. Let S be a set of functional dependencies on schema R , and let $R_1, R_2, \dots, R_n$ be the result of a 3NF decomposition of R . Let S_i be a minimal basis for R_i , and define $S_{\text{union}} = S_1 \cup S_2 \cup \dots \cup S_n$. Then for every functional dependency $\alpha \rightarrow \beta$ in S , it holds that $\alpha \rightarrow \beta$ follows from S_{union} .
3. If a 3-column relation R violates BCNF, then the result of performing one iteration of the BCNF decomposition algorithm is always in BCNF.
4. In set-based relational algebra, it is possible to create expressions that are equivalent to a natural join ($\bowtie$) by using left-outer-join ($\bowtie\!\!\!\!\!\triangleright$), union ($\cup$), intersection ($\cap$), and projection (π).

Problem 3 (9 points)

All of the following statements are false. Demonstrate the falsehood of each statement by performing the indicated action. No partial points will be given for each subquestion.

(1) (3 points)

Statement: A basis B that satisfies the following two conditions will be considered a minimal basis:

1. If any functional dependency is removed from B , the result is no longer a basis.
2. If one or more attributes are removed from the left-hand side of any functional dependency in B , the result is no longer a basis.

According to the conditions specified above, the set $S = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, AD \rightarrow B\}$ qualifies as a minimal basis.

Action: Give one minimal basis equivalent to S that satisfies the above conditions.

(2) (3 points)

Statement: If $\alpha \rightarrow \gamma$ and $\beta \rightarrow \gamma$, then at least one of $\alpha \rightarrow \beta$ and $\beta \rightarrow \alpha$ always holds.

Action: Give one possible counterexample.

(3) (3 points)

Statement: The multivalued dependency $A \twoheadrightarrow B$ holds in R .

Action: Add proper tuples to R to make $A \twoheadrightarrow B$ hold.

A	B	C
0	0	2
0	1	3
0	4	2

Problem 4 (12 points)

Consider the following set S of functional dependencies on the relation schema $R(A, B, C, D, E, F)$:

$$S = \{AB \rightarrow C, CD \rightarrow E, E \rightarrow F, F \rightarrow D\}$$

(1) Compute the closure $\{CF\}^+$ with respect to S . You do not need to write the solution process. **(2 points)**

(2) List all candidate keys of R . You do not need to write the solution process. **(3 points)**

(3) Suppose we project out attributes A and B , resulting in the relation $R_1(C, D, E, F)$. Compute a minimal basis for the functional dependencies of R_1 . The conditions for a minimal basis are the same as in Problem 2(1). If multiple minimal bases exist, providing one possible minimal basis is sufficient. **(3 points)**

(4) Are there any BCNF violations in R_1 defined in (3)? If so, decompose the relation into BCNF using the BCNF decomposition algorithm. Use the minimal basis set obtained in (3) as the set of functional dependencies of R_1 when performing the BCNF decomposition. Follow the BCNF decomposition algorithm covered in the class. **(4 points)**

Problem 5 (10 points)

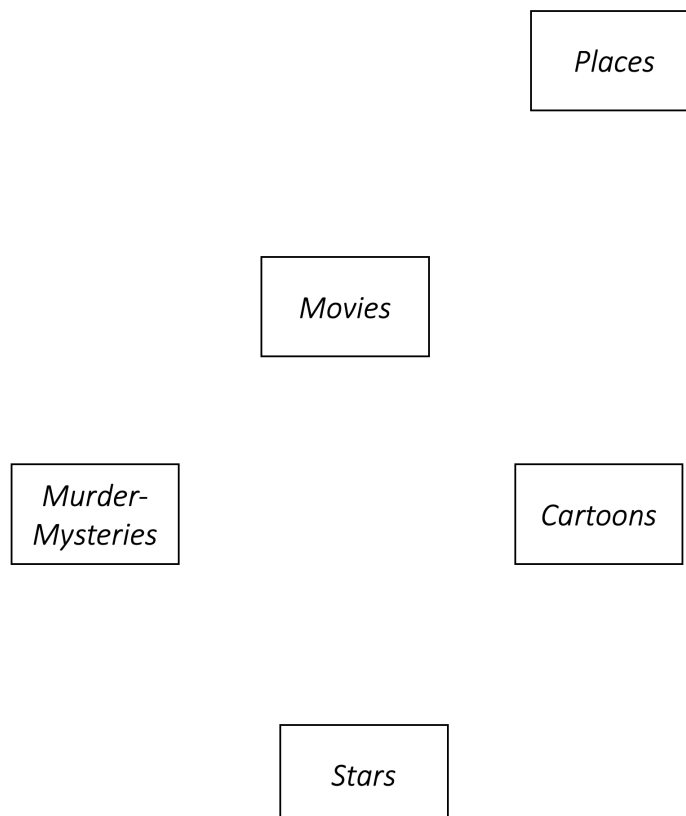
You are building a movie information management database. The following describes the system requirements:

- Each *Movie* is filmed at exactly one *Place*. *Places* have a city.
- A *Cartoon* movie involves multiple *Stars*, and for each *Star*, the character name they voice must be recorded.
- A *MurderMystery* movie features multiple *Stars*, and for each *Star*, the number of minutes they appear must be recorded.
- Both *Cartoons* and *MurderMysteries* are subclasses of *Movies*.
- A *Movie* can have at most one premiere held at a *Place*. The premiere is optional and may not exist for every movie.

Using only the given entity sets, complete an **Entity-Relationship (ER) diagram** that satisfies all the requirements above. Your diagram must include:

- All necessary relationships, including participating entity sets and cardinalities (1:1, 1:N, M:N)
- Any attributes on relationships and entity sets, if applicable
- Subclass relationships explicitly shown

The names of relationships will not be graded, but their structure must be clearly indicated. Include only the content in the requirements, and **follow the ER modeling conventions covered in the class**.



Problem 6 (8 points)

The safety of a Datalog is crucial to ensure the termination and computability of query execution.

Determine the safety of each of the data rules below. If a rule is safe, write a relational algebra expression that produces the same result as the head of the rule.

$$(1) A(x, y, z) \leftarrow NOT B(x, y) AND NOT C(y, z) AND D(x, z)$$

$$(2) P(a, b) \leftarrow R(a, b) AND S(b, c) AND c \geq 10$$

$$(3) A(u, v, w) \leftarrow B(u, w) AND C(w, v) AND D(v)$$

Problem 7 (10 points)

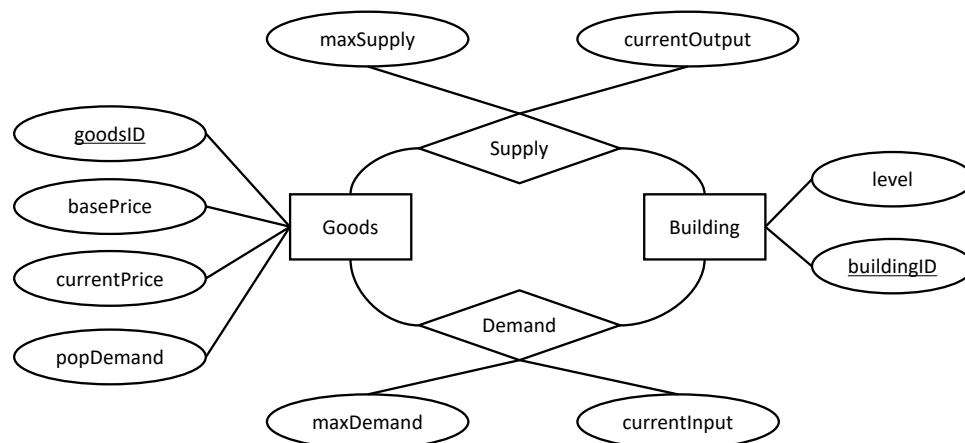
In this problem, your goal is to design and use a database for managing a supply chain. Following table shows the explanations for each attributes. Here, underline indicates the key of the database.

Attribute name	Meaning
<u>goodsID</u>	goods ID
<u>buildingID</u>	building ID
maxSupply	Maximum amount of the goods that could be produced by the building
currentOutput	Current amount of the goods produced by the building
maxDemand	Maximum amount of the goods that could be consumed by the building
currentInput	Current amount of the goods consumed by the building
basePrice	Base price of goods
currentPrice	Current price of goods
popDemand	Minimum amount of demand for the goods
level	Size of the building

(1) Based on the meaning of attributes, list all the functional dependencies in the schema from the table. Your functional dependency set should satisfy minimal basis conditions as explained in Problem 3(1). Note that some attributes are omitted. **(6 points)**

goodsID	buildingID	level	currentOutput	maxSupply	maxDemand	currentPrice	popDemand
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(2) Based on (1) and the explanation, assume that we construct an ER diagram. Define the relational schema from the **given ER diagram**. You should underline the key attributes in each table. **(4 points)**



Problem 8 (7 points)

Following figure shows a RDB schema for an international trading scenario, in two different forms: (a) a non-normalized table and (b) normalized tables. Your goal in this problem is to write SQL queries that calculate and utilize the TPR , defined as:

$$TPR(cid, nid) = \frac{TP_country(cid, nid)}{TP_total(nid)}$$

(a)

locating

<u>cid</u>	<u>nid</u>	home	dev	LV	IV	TP-total	TP-country
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(b)

country

<u>cid</u>	home	dev
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trading_node

<u>nid</u>	LV	IV	TP-total
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node_country

<u>cid</u>	<u>nid</u>	TP-country
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Figure 1: The RDB schema for (a) non-normalized and (b) normalized table. **Bold** indicates the name of the table.

(1) Write the SQL using schema (a) to calculate TPR where the `cid` is `POR` and the `nid` is `sevilla`.
(3 points)

(2) Write the SQL using schema (b) to calculate TPR where the `cid` is `POR` and the `nid` is `sevilla`.
(3 points)

(3) Recently, there is a growing interest in using language models (LMs, e.g., ChatGPT) to generate SQL queries that can answer questions like (1) and (2). Many studies have reported that the difficulty of those generations increase in proportion to the number of `JOIN`, `GROUP BY`, and `ORDER BY` clauses in the query to be generated (i.e., if there are more `JOINS`, query generation becomes much harder). From this, when building a system that combines an LM with a relational database, **which format—(a) or (b)—would be useful, in terms of the inference performance of the LM?** You do not need to justify your answer. (1 points)

Problem 9 (13 points)

Consider the relational algebra on ***bags*** (not sets) with the following relational operators: ***Bag Union**($\cup$), **Bag Intersection**($\cap$), **Bag Difference**($-$), **Projection**(π), **Natural Join**($\bowtie$)*. Fill in the blanks below, no explanation is needed as there are no partial scores.

Enrollment		Courses	
<i>studentName</i> (char 100)	<i>courseCode</i> (char 10)	<i>courseCode</i> (char 10)	<i>courseName</i> (char 1000)
James	CS204	CS204	Discrete Mathematics
James	CS360	CS360	Databases (section A)
Paul	EE210	CS360	Databases (section B)
Sarah	EE312	CS376	Machine Learning

Full size of Enrollment = # tuple * size of tuple = 4 * (100 + 10) = 440

Full size of Courses = # tuple * size of tuple = 4 * (10 + 1000) = 4040

Figure 2: Example tables.

Enrollment $\bowtie$ Courses		Enrollment $\triangleright$ Courses	
<i>studentName</i>	<i>courseCode</i>	<i>studentName</i>	<i>courseCode</i>
James	CS204	Paul	EE210
James	CS360	Sarah	EE312

(a) semi-join results (b) anti-join results

Figure 3: Example query results.

(1) The semi-join of $R(A, B)$ and $S(B, C)$, denoted as $R \bowtie S$, is the set of tuples in R that can successfully natural join with at least one tuple in S . Figure 3(a) shows the example of a semi-join on the tables in Figure 2. Using only the operators mentioned in this problem, give one algebraic expression, which is equivalent to $R(A, B) \bowtie S(B, C)$. (3 points)

$R(A, B) \bowtie S(B, C) =$ _____

(2) The anti-join of $R(A, B)$ and $S(B, C)$, denoted as $R \triangleright S$, is the set of tuples in R that do not natural join with any tuple in S . Figure 3(b) shows the example of anti-join on the tables in Figure 2. Using only the operators mentioned in this problem, give one algebraic expression, which is equivalent to $R(A, B) \triangleright S(B, C)$. (3 points)

$R(A, B) \triangleright S(B, C) =$ _____

(3) Imagine that Enrollment and Courses in Figure 2 are stored on different servers A and B, respectively. Then, to natural join the two tables, we need to fetch data from the other server in one direction or another (we named these operators *fetchA* and *fetchB*). So, we have two possible algebraic expressions: 1) $\text{fetchA}(E) \bowtie C$, 2) $E \bowtie \text{fetchB}(C)$ (we use E and C as shorthand for Enrollment and Courses, respectively). In this case, the amount of data fetched in the first expression is 440 bytes, the full size of the Enrollment table, and the amount of data fetched in the second expression is 4040 bytes, the full size of the Courses table (see Figure 2 for how we got the size). Therefore, the first expression is more efficient. Using only the operators mentioned in this problem (in here, *fetchA*, *fetchB*, $\bowtie$ and $\triangleright$ can also be used), give one expression that is equivalent to the natural join result that best minimizes the amount of fetches. (5 points)

$\text{fetchA}(E) \bowtie C =$ _____

(4) Write down the minimum fetch amount in (3) (2 points)

Problem 10 (16 points)

Consider the relational algebra on ***bags*** (not sets) with the following relational operators: ***Cross Product** ($\times$), **Natural Join** ($\bowtie$), **Semi Join** ($\ltimes$), **Anti Join** ($\bar{\bowtie}$), **Left Outer Join** ($\ltimes\ltimes$), **Full Outer Join** ($\ltimes\ltimes\ltimes$) * (Semi Join and Anti Join are defined in Problem 9).

(1) We say that a binary algebraic operator (denoted as $\circ$) is commutative if the following equivalence holds: $R \circ S = S \circ R$. The table below indicates whether each operation is commutative. For example, natural join is commutative and is therefore filled with $\checkmark$. Fill in the rest of the table by marking Yes ($\checkmark$) or No (X). No explanation is needed.

(4 points, 1 point for each correct, -1 point for each incorrect)

$\circ$				
$\bowtie$	$\ltimes$	$\bar{\bowtie}$	$\ltimes\ltimes$	$\ltimes\ltimes\ltimes$
$\checkmark$				

(2) We say that two binary algebraic operators (denoted as $\circ^a$, $\circ^b$) are associative if the following equivalence holds: $(R \circ^a S) \circ^b T = R \circ^a (S \circ^b T)$. The table below indicates whether each pair of operations $\circ^a$ (row) combined with $\circ^b$ (column) is associative. For example, pair of cross products is associative and is therefore filled with $\checkmark$. Fill in the rest of the table by marking Yes ($\checkmark$) or No (X). No explanation is needed.

(4 points, 0.5 points for each correct, -0.5 points for each incorrect)

$\circ^a$	$\circ^b$		
	$\times$	$\bowtie$	$\ltimes$
$\times$	$\checkmark$		
$\bowtie$			
$\ltimes$			

(3) The database system has a module called *search space enumerator*. This module generates all the equivalent algebraic expressions based on the algebraic properties we defined above (e.g., commutativity, associativity) and passes them to the *query optimizer* module to help it choose the most efficient one (*learn more about this in the remaining lectures* ☺). This time, become a human search space enumerator and generate all possible expressions that are equivalent to the expression below. To prevent the search space from becoming too large, there are two constraints: 1) use only given operators ($\bowtie$, $\ltimes$), and 2) use R , S , and T only once within an expression. Fill in the dotted blanks as you find them without explanation. (8 points, -1 point for each incorrect)

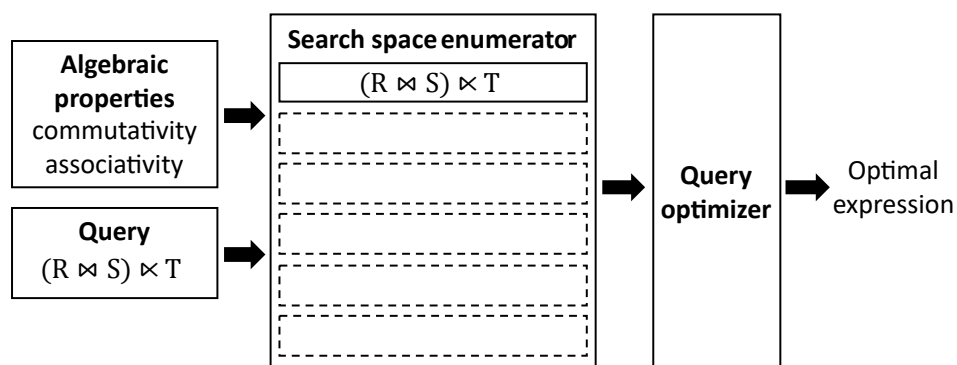


Figure 4: Query optimization flow.